D15 JustVibeAndTapeout

MOSAIC-SOC: Agent-Driven, Reconfigurable Multi-Core SoC Generator using Open-Source EDA

Team members

Discord name	Affiliation	Role	Experience
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1. Project Information

INFO

TRACK D

MOSAIC-SoC

MOSAIC-SoC extends X-HEEP into a configuration-driven heterogeneous RISC-V SoC. One validated mosaic.yaml selects the core mix and generates the OBI fabric, logical memory map, peripherals, and downstream flow inputs.

Latest repository additions

HARNESS	oh-my-soc CLI plus four deterministic skills.
FIRMWARE	Bare-metal TDU control, worker programs, and three scheduling modes.
MEMORY	GF180 OpenRAM/LibreLane scaffolding; characterized SRAM macros remain open.

Review baseline · HEAD d5cae92 · 2 July 2026 · DASHBOARD claims cross-checked against code

Team

@MILOUDIAS	Lead
@trabdelbasset	Member
@yacine-hk	Member

Current PoC

1× TITAN · 2× ATLAS · 4× NANO

7	**32 KB**	**50**
harts	logical SRAM	MHz

50 MHz review target
TDU · iDMA · generated OBI

2. Phase Plan & Progress

INFO

PHASE 1

95% · WORKING

Config-driven multi-core generator

Goal: one validated mosaic.yaml selects cores, memory, bus fabric, scheduler, and peripherals, and generates the heterogeneous SoC from X-HEEP (PoC: 1× TITAN + 2× ATLAS + 4× NANO).

- SCI wrappers for all 5 core IPs (cv32e20, Ibex, FazyRV, QERV, SERV).
- TDU + iDMA integrated; 4 testbench suites PASS; 14 bugs fixed.
- Full-SoC lint-clean (837 modules); 3-core wake demo EXIT SUCCESS.

PHASE 2

85% · IMPLEMENTED

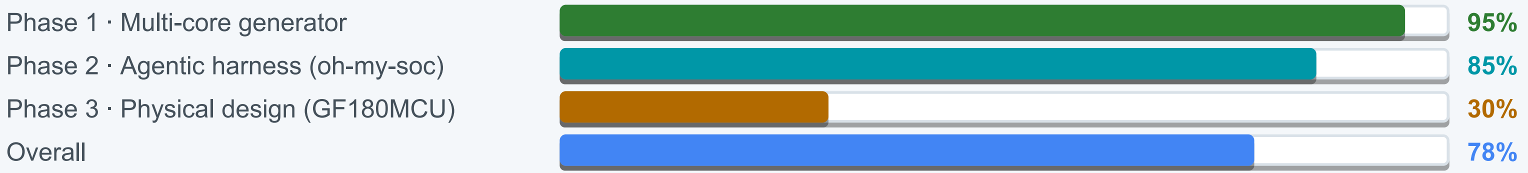
Agentic harness — oh-my-soc

Goal: an agentic harness (based on oh-my-pi) that automates config authoring, EDA flow running, DRC triage, and documentation. Deterministic skills check the agent; it never replaces signoff.

- 4 skills + CLI: config-author (3 presets), flow-runner (11 flows), drc-triage (3 report parsers), doc-gen.
- Structured SkillResult JSON; manual make/CLI fallback preserved.
- Open: harness regression suite and flow-runner path handling.

DASHBOARD.md · Executive summary — 1 July 2026

Milestones: M1–M9 DONE (Jun 27–30) · M10 pin-binding + SRAM IN PROGRESS · M11–M12 signoff / GDSII PLANNED



3. Tapeout Plan & Long-Term Goal

INFO

The PoC exercises the flow end-to-end; the 2026 die is a minimal configuration sized to the area budget

2026 TAPEOUT

Minimal die: 1× FazyRV + 2× SERV

- 3 harts : FazyRV-CHUNK8 as the boot/orchestrator hart; 2× SERV workers woken through the TDU.
- No on-chip SRAM macros: external memory only, to meet the GF180 area constraints.
- Tiny logic: FazyRV ≈2–5 kGE, SERV ≈0.2–3 kGE per core (vs cv32e20 ≈14–17 kGE plus SRAM macros).
- Generated by the same mosaic.yaml flow; the tapeout config variant is to be frozen at this review.

CURRENT PoC

Flow validation — not the 2026 die

- The 7-hart PoC (1× cv32e20 + 2× FazyRV + 4× SERV, 32 KB SRAM) exists to test the flow itself: generator → firmware → TDU → verification.
- It stays the simulation regression vehicle and is not intended for the Chipathon 2026 tapeout.
- On-chip SRAM physicalization (OpenRAM) moves off the 2026 critical path and remains open for future runs.

LONG-TERM GOAL

Grow MOSAIC into a broad IP catalog

Integrate as many core and peripheral IPs as possible behind the SCI/OBI contract — adding a core is one SCI wrapper, one .core descriptor, and one template branch. Today: 5 core IPs (cv32e20, Ibex, FazyRV, QERV, SERV) and 4 peripherals (UART, GPIO, timer, SPI). The generator sizes the OBI fabric and memory map automatically as the catalog grows, so future dies scale up as area allows.

4. Design Objectives

INFO

The schematic must support a useful PoC and stay reproducible from one YAML or handwritten or LLM-authored

01 Configuration-driven

Generate topology, memory map, firmware assumptions, and flow inputs from one validated YAML.

02 Heterogeneous execution

TITAN dispatches work; ATLAS and NANO workers start dormant and wake through the TDU.

03 Firmware control plane

Provide an RV32I TDU driver, per-tier worker code, and static/dynamic/power-aware scheduling demos.

04 Deterministic harness

Route agent actions through validated skills that return structured results and preserve manual fallback.

05 Physicalized memory

Replace generic SRAM with characterized GF180 macros and explicit LibreLane placement/PDN integration.

06 Evidence-gated tapeout

Require reproducible RTL, firmware, STA, DRC/LVS, antenna, and review artifacts before signoff.

5. System Overview

INFO

mosaic.yaml
cores · SRAM · I/O

MOSAIC
validate · generate

Generated heterogeneous core cluster

- **TITAN** — 1× cv32e20
- **ATLAS** — 2× FazyRV-CHUNK8
- **NANO** — 4× SERV

Standard Core Interface wrappers
OBI I/D · hart ID · wake · clock-stall adapter

Parameterized OBI N×M system interconnect

CONTROL + DATA MOVEMENT
TDU · wake / FIFO / CPI-energy counters
iDMA · register FE / ND / native OBI

MEMORY SYSTEM
32 KB logical banked SRAM · 2 KB boot ROM
GF180 macro substitution remains open

PLATFORM SERVICES + I/O
Debug · PLIC · timers · power manager
UART · GPIO · SPI

6. oh-my-soc Agentic Harness

HARNESS

The AI/LLM contribution (Phase 2): deterministic Python skills around configuration, flows, reports, and documentation

Agent / human
intent · config · report

oh-my-soc CLI
python -m harness

4 SKILLS

SkillResult JSON
ok · metrics · errors

Deterministic tools
make · Verilator · LibreLane

CONFIG-AUTHOR
3 presets · schema validation
valid mosaic.yaml output

FLOW-RUNNER
11 registered flows · timeout
structured log summaries

DRC-TRIAGE
Magic/KLayout/Netgen parsers
classification + suggestions

DOC-GEN
config · memory map · run report
DASHBOARD extraction

Proposal-review response: no specially trained LLM is needed, module connection is generated deterministically from the schema-validated mosaic.yaml; the agent authors, runs, triages, and documents.

Verified now: config validation, preset/list commands, memory-map and dashboard generation. **Still open:** complete harness regression and path-safe firmware flow execution.

7. Core, Interconnect & Scheduling Arch

SYSTEM

A tiered execution model over one generated memory map and one shared fabric

T1 · TITAN — **1× cv32e20**

Boots immediately
Owns external interrupts
Dispatches worker tasks

T2 · ATLAS — **2× FazyRV-CHUNK8**

Chunk-serial workers
Dormant at reset
Wake through TDU

T3 · NANO — **4× SERV**

Bit-serial workers
Dormant at reset
Wake through TDU

SCI wrappers normalize each core to OBI 1.3

2 masters per hart + debug + iDMA ports → generated OBI crossbar

TDU @ 0x200A0000

banked SRAM + ROM

peripherals + debug

8. Firmware & Scheduling Modes

FIRMWARE

New bare-metal RV32I firmware programs the TDU and provides per-tier worker workloads

TITAN control plane

cv32e20 @ 0x180

- tdu.{h,c}: mode, wake mask, task FIFO, CPI estimates, energy counter.
- titan_main.c dispatches ATLAS and NANO work and polls completion sentinels.
- titan_scheduling_demo.c runs STATIC, DYNAMIC, and POWER_AWARE phases.
- Current implementation is bare-metal; FreeRTOS task integration remains future work.

PASS **Build verified**

ATLAS

FazyRV @ 0x1000

RV32I assembly
shift/add MAC
result + sentinel

NANO

SERV @ 0x2000

RV32I assembly
sensor/CRC
result + sentinel

Build evidence

1492 B	2348 B
main text	demo text

ELF + HEX built
RV32 GCC

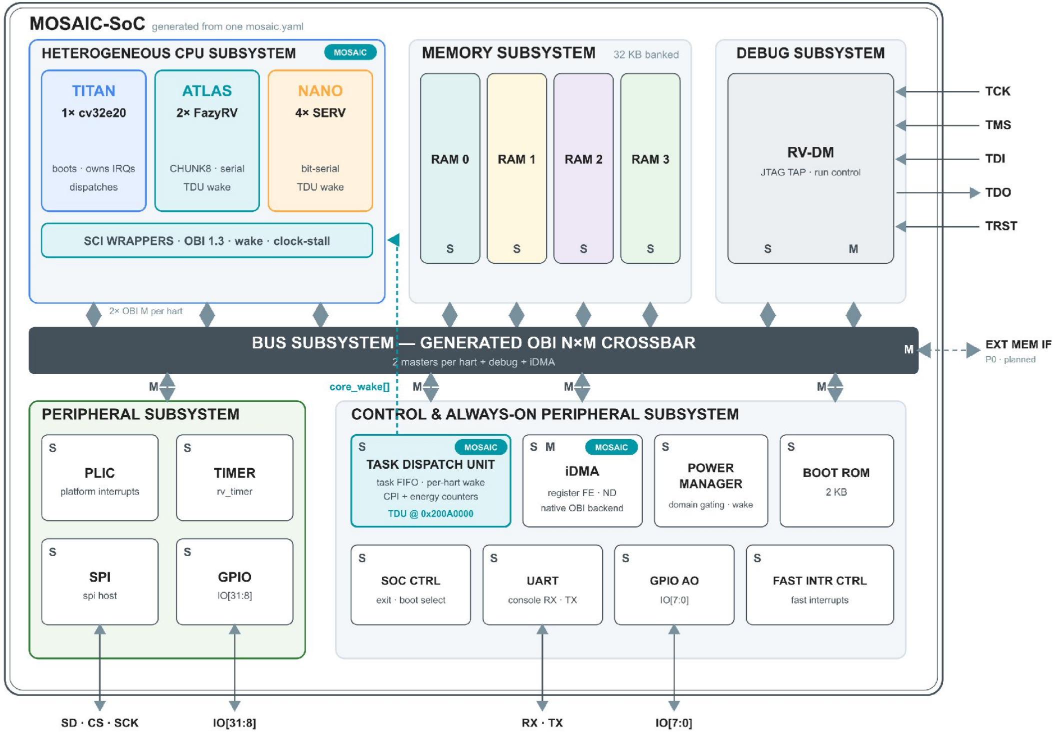
Scheduling demonstration

STATIC	DYNAMIC	POWER AWARE
fixed target assignment	CPI-based core selection	energy-budget consolidation

9. Schematic Summary — Architecture

SCHEMATIC

Top-level block diagram adapted from the X-HEEP diagram; MOSAIC additions are tagged : PoC configuration shown



PoC configuration shown — 7 harts (1× cv32e20 · 2× FazyRV · 4× SERV) · 32 KB SRAM · 50 MHz

2026 tapeout die: 1× FazyRV + 2× SERV, no SRAM banks — external memory via EXT MEM IF (definition open, P0)

Reading the diagram

- X-HEEP skeleton kept: memory, debug, peripheral and AO subsystems on one bus; S/M port convention.
- MOSAIC additions tagged: heterogeneous tiers behind SCI wrappers, TDU @ 0x200A0000, iDMA.
- Dashed teal core_wake[]: the TDU wakes dormant ATLAS / NANO harts through the SCI wrappers.
- EXT MEM IF is dashed: the off-chip memory port for the Chipathon tapeout

9. Schematic Summary — Hierarchy

SCHEMATIC

Hierarchical RTL view and the principal interface contracts that require review

Hierarchy / block	Function	Primary interfaces	Current source
core_v_mini_mcu	Top-level MCU + generated multicore subsystem	clk/reset · pads · OBI · interrupts	hw/core-v-mini-mcu/*.sv.tpl
cpu_subsystem	Instantiates all configured harts and SCI wrappers	OBI I/D per hart · hart ID · wake	cpu_subsystem.sv.tpl
SCI wrappers	Core-bus conversion and dormancy/wake behavior	Wishbone/req-gnt ↔ OBI 1.3	hw/sci/
system_bus	Generated multi-master address decode and arbitration	OBI N×M	core_v_mini_mcu_pkg.sv.tpl
TDU	Task FIFO, per-hart wake, counters	AO register bus · core_wake[]	hw/tdu/
iDMA	Memory-to-memory transfer engine	register FE · native OBI backend	hw/vendor/mosaic/idma/
memory / OpenRAM	Logical banks plus pending GF180 macro replacement	OBI bank ports · VDD/GND views	sw/vendor/openram/ + flow/librelane/
chip wrapper	GF180 pads and SoC pin adapter	pad buses · VDD/VSS · clk/reset	flow/librelane/src/

10. OpenRAM & SRAM Physicalization

MEMORY

GF180 tech/config code and LibreLane hooks exist; macros not generated , the 2026 die uses external memory

1 **GF180 port code**
tech.py · 3.3 V
layers · DRC estimates

2 **Macro configs**
4k.py / 32k.py
configs added

3 **Macro outputs**
GDS / LEF / LIB /
SPICE absent

4 **LibreLane hooks**
MACROS · placement
PDN grid / connections

5 **Mapped signoff**
binding · STA
DRC / LVS / antenna

Capacity audit

File / label	Configured geometry	Actual capacity
mosaic_sram_4k.py	512 × 8 bits	512 B
mosaic_sram_32k.py	4096 × 8 bits	4 KB
Required 4 KB	4096 × 8 bits	4 KB
Required 32 KB	32768 × 8 bits	32 KB

On-chip SRAM: close before a future run

- Correct the byte-capacity configurations and choose the future on-chip memory size.
- Obtain/validate bitcell and peripheral cells; generate characterized GDS/LEF/LIB/SPICE/Verilog.
- Align GND/VSS naming, macro instance binding, placement, PDN, and RTL memory wrapper.
- Re-run area and timing; current 0.05 / 0.3-0.6 mm² values are estimates only.

11. Design Assumptions

ASSUME

Known constraints are separated from items that still require confirmation

CLOCK & RESET

• One external pad clock; current SDC target is 50 MHz / 20 ns.
• Worker-core clocks may stall or gate during dormancy and outstanding serial-core transactions.
• Derived/gated-clock constraints and reset-domain checks are not yet signoff-complete.

MEMORY & BUS

• 32 KB logical SRAM plus 2 KB boot ROM; bank count and addresses are generated.
• OBI 1.3 is the internal contract; SCI wrappers absorb native core-bus differences.
• OpenRAM files are port/config scaffolding only;

POWER & I/O

• Current chip flow exposes VDD/VSS; OpenRAM code uses VDD/GND and requires pin-name reconciliation.
• TITAN boots; ATLAS/NANO workers start dormant and wake via TDU.
• Final I/O voltage/domain assignment, pad order, bonding map, and pad-to-SoC binding need review.

VERIFICATION BOUNDARY

• Simulation proves selected functional paths; it is not physical signoff.
• New firmware builds but has not replaced the legacy assembly program in the full-SoC test.
• The harness is implemented, but deterministic EDA outputs still require independent tool evidence.

12. Verification Status

VERIFY

Latest software additions are separated into build/smoke evidence versus integrated execution and physical signoff

Verification item	Evidence	Status
Harness config/doc	validate, presets, flow list, dashboard and memory-map commands	PASS
Firmware build	RV32I main/demo ELF+HEX; 1492 B / 2348 B text	PASS
Full-SoC elaboration	Verilator lint-only; repository reports 837 modules	PASS
3-core wake-and-run	cv32e20 wakes FazyRV + SERV; both write sentinels; EXIT SUCCESS	PASS
TDU + iDMA + SCI	Existing unit/SoC/cocotb regressions report PASS	PASS
Harness flow execution	firmware flow fails under current space-containing repo path	PARTIAL
OpenRAM macros	Port/config/hooks exist; no GDS/LEF/LIB/SPICE outputs	OPEN
GF180 signoff	Pin binding, mapped SRAM, STA, DRC/LVS/antenna incomplete	OPEN

EVIDENCE

LOG
tb/mosaic_soc/sim.log

FW
sw/firmware/build/*.elf

HRN
python -m harness

RAM
sw/vendor/openram

13. Design Checklist

CHECK

Required review categories mapped to current evidence and remaining actions

Functionality

PARTIAL

Legacy wake path passes; integrate the new firmware HEX and full 7-core workload.

Analog

N/A

Digital SoC; no custom analog IP. Review pad-library use and supply assumptions.

Digital

PARTIAL

RTL/harness/firmware code exists. Correct SRAM capacity, bind pins/macros, then synthesize and sign off.

Mixed Signal

N/A

No mixed-signal block. Confirm digital pad models and off-chip interface assumptions.

Reliability

PARTIAL

Reset/wake tested. CDC/RDC, gated clocks, SRAM corners, antenna, IR/EM and waivers remain.

Documentation

PARTIAL

README/DASHBOARD/harness docs exist; reconcile stale metrics and attach final reports.

14. Open Issues & Risk-Retirement Plan

RISKS

Pri.	Issue	Required action / closure evidence
P0	External-memory path undefined	Define and implement the off-chip memory interface (pads, controller, boot path) for the tapeout die.
P0	SoC pin binding incomplete	Instantiate x_heep_system in mosaic_soc_core.sv; map every pad bit; lint + connectivity review.
P0	No complete signoff run	Finish flatten/slang path; produce synthesis, STA, DRC, LVS, antenna reports and final GDS.
P1	New firmware not integrated	Load mosaic_fw/demo.hex in the full-SoC TB and prove all configured worker harts complete.
P1	Harness regression incomplete	Add unit/integration tests; fix space-containing path handling and flow-runner edge cases.
P1	Pad / clock intent not frozen	Finalize pad order, bonding, derived/gated clocks, resets, CDC/RDC and timing exceptions.
P1	DRC waiver posture inherited	Justify every waiver and restore error-on-violation behavior for tapeout acceptance.
P1	OpenRAM capacity is wrong	Fix 512×8 / 4096×8 byte-capacity labels; on-chip SRAM is future-run scope (2026 die is external-memory).
P1	SRAM macro outputs absent	Future-run scope: generate/characterize GDS/LEF/LIB/SPICE, bind instance, align GND/VSS + PDN.

15. Questions for Reviewers

REVIEW

Decisions requested now to prevent rework before the physical-design run

01 Memory plan

Approve external-memory-only for the 2026 die and select the off-chip interface; corrected on-chip SRAM is deferred.

02 Tapeout configuration

Approve the minimal 1× FazyRV + 2× SERV die, with the 7-hart PoC remaining the flow-validation vehicle only?

03 Firmware scope

Is bare-metal TDU scheduling sufficient for review, or is FreeRTOS integration required before freeze?

04 Harness acceptance

Which harness commands and failure-recovery cases must be demonstrated as the Track D deliverable?

05 Clock, power and pads

Are 50 MHz, the clock-stall semantics, and a pad budget that includes the external-memory interface acceptable?

06 Signoff policy

Which waivers are acceptable, and what STA/DRC/LVS/antenna evidence package closes the review?

16. Review Outcome

REVIEW

Requested disposition

- ☐ Approve schematic freeze
- ☐ Approve with actions
- ☐ Re-review required

PENDING

Reviewer notes / decision:

Proposed exit criteria

1. **Configuration frozen**
   Tapeout core mix (1× FazyRV + 2× SERV), external-memory plan, clock/power target and pad budget approved.
2. **Firmware integrated**
   Final HEX boots in full-SoC simulation and all selected worker harts complete.
3. **Harness evidenced**
   Required skills and failure paths return reproducible structured results.
4. **Memory plan closed**
   External-memory interface defined and simulated; on-chip SRAM macros tracked as future work.
5. **Signoff complete**
   STA, DRC/LVS, antenna and justified waivers are archived and approved.

Appendix A · PoC Config (Flow Validation)

APPX

Current mosaic.yaml PoC exercises the flow; the 2026 tapeout die is the minimal FazyRV + 2× SERV configuration


```yaml
soc:
  name: mosaic_poc_alpha
  pdk: gf180mcu
  cores:
    - {ip: cv32e20, count: 1, role: titan,
isa: rv32emc}
    - {ip: fazyrv, count: 2, role: atlas,
chunksize: 8}
    - {ip: serv, count: 4, role: nano,
isa: rv32i}
  memory: {sram_kb: 32, boot_rom_kb: 2}
  bus: obi
  scheduler: {tdu: true, mode: dynamic}
  peripherals: [uart, gpio, timer, spi]
```


Harness-validated topology

mosaic.yaml validates: 7 harts
4 peripherals · dynamic TDU

Memory configuration conflict

YAML requests 32 KB logical SRAM
LibreLane defaults to '4k' macro
whose config is actually 512 B

Current integrated demo

mosaic_wake_demo.yaml: 1× cv32e20 +
1× FazyRV + 1× SERV · current full-SoC demo

Appendix B · Verification Evidence

APPX

Commands and artifacts to place in the schematic-review package

HARNESS	`python -m harness config-author validate mosaic.yaml`	PASS · 7 cores
FIRMWARE	`make ROOT=/tmp/mosaic_soc_latest RISCV_TC=riscv32-unknown-elf`	ELF + HEX build
FULL-SOC DEMO	`tb/mosaic_soc/run.sh`	legacy fw: EXIT SUCCESS
REGRESSIONS	`tb/mosaic/cocotb/run.sh; tb/tdu/...; tb/idma/...`	existing PASS evidence
OPENRAM	`python sram_compiler.py <corrected_config.py>`	PENDING macro views
HARDENING	`cd flow/librelane && make harden SLOT=mosaic`	PENDING signoff